WORKED SOLUTION

STEP III 2018, Question 9

Mechanics • Collisions and Recurrence Relations

STEP III 2018, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q9

The key idea

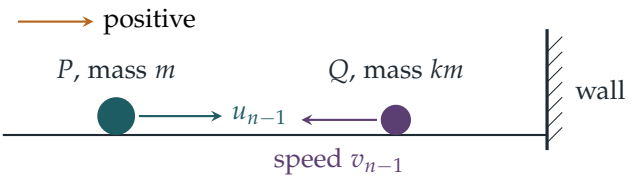
Each collision between P and Q gives two linear equations, one from conservation of momentum and one from Newton's law of restitution, linking the velocities before and after. The wall simply reverses Q's velocity, so the state before one collision is fixed by the state after the previous one. That gives a pair of first-order recurrences for u_n and v_n, and eliminating v leaves a single second-order recurrence for u_n. Part (ii) is then the usual business of fitting a solution of a linear recurrence to its first two terms.

Solution

We model P and Q as particles moving along one line on a smooth floor, so no horizontal forces act between collisions and each particle keeps a constant velocity. Take the direction towards the wall as positive throughout.

What happens between collisions. After the nth collision Q has velocity v_n (towards the wall, since it collides with the wall next). The wall collision is perfectly elastic, so Q comes back with velocity $-v_n$, while P still has velocity u_n. So just before collision $n+1$ the velocities are u_n (for P) and $-v_n$ (for Q).

This also holds for the first collision if we read v_0 the right way: before it, Q moves away from the wall with speed v_0, so its velocity is $-v_0$, exactly the pattern above with $n = 0$.



Part (i)

For the nth collision ($n \geq 1$), conservation of momentum (dividing through by m) gives

$$u_{n-1} - kv_{n-1} = u_n + kv_n. \tag{1}$$

The speed of approach is $u_{n-1} - (-v_{n-1}) = u_{n-1} + v_{n-1}$. Afterwards Q, which is nearer the wall, moves towards the wall faster than P, so the speed of separation is $v_n - u_n$. Newton's law of restitution gives

$$v_n - u_n = e\,(u_{n-1} + v_{n-1}). \tag{2}$$

Substituting v_n from (2) into (1):

$$u_{n-1} - kv_{n-1} = (1+k)u_n + ke\,(u_{n-1} + v_{n-1}),$$

so

$$k(1+e)\,v_{n-1} = (1-ke)\,u_{n-1} - (1+k)\,u_n \qquad (n \geq 1). \tag{3}$$

This expresses each v in terms of two consecutive u's, which is exactly what we need to eliminate v.

Ask yourself Equation (3) gets rid of v_{n-1}, but (2) also contains v_n. How do we deal with that?

Equation (3) holds for every $n \geq 1$, so we may also use it with $n+1$ in place of n:

$$k(1+e)\,v_n = (1-ke)\,u_n - (1+k)\,u_{n+1}.$$

Now multiply (2) by $k(1+e)$ and replace both $k(1+e)v_n$ and $k(1+e)v_{n-1}$ using these:

$$(1-ke)u_n - (1+k)u_{n+1} - k(1+e)u_n = ke(1+e)\,u_{n-1} + e[(1-ke)u_{n-1} - (1+k)u_n].$$

Collect terms. The coefficient of u_{n-1} on the right is $ke + ke^2 + e - ke^2 = e(1+k)$. Moving the u_n terms to the left, their coefficient is

$$1 - ke - k - ke + e + ke = 1 - k + e - ke = (1+e)(1-k).$$

Hence

$$(1+k)u_{n+1} - (1-k)(1+e)u_n + e(1+k)u_{n-1} = 0 \qquad (n \geq 1),$$

and replacing n by $n-1$ gives $(*)$ for $n \geq 2$, as required.

Part (ii)

With $e = \frac{1}{2}$ and $k = \frac{1}{34}$, multiplying $(*)$ by 68 gives

$$70u_n - 99u_{n-1} + 35u_{n-2} = 0,$$

whose auxiliary equation $70t^2 - 99t + 35 = (10t-7)(7t-5) = 0$ has roots $\frac{7}{10}$ and $\frac{5}{7}$, matching the given form.

Two conditions fix A and B, and we need the first two terms. We know u_0. For u_1, use (3) with $n = 1$:

$$(1+k)u_1 = (1-ke)u_0 - k(1+e)v_0,$$
$$\text{so} \quad \tfrac{35}{34}u_1 = \tfrac{67}{68}u_0 - \tfrac{3}{68}v_0, \quad \text{giving} \quad u_1 = \frac{67u_0 - 3v_0}{70}.$$

Putting $n = 0$ and $n = 1$ into the given form,

$$A + B = u_0, \qquad \tfrac{7}{10}A + \tfrac{5}{7}B = \frac{67u_0 - 3v_0}{70}.$$

Multiplying the second equation by 70 gives $49A + 50B = 67u_0 - 3v_0$. Subtracting 49 times the first,

$$B = 18u_0 - 3v_0, \qquad A = u_0 - B = 3v_0 - 17u_0.$$

Sign of u_n for large n. Since $\frac{5}{7} > \frac{7}{10}$ (as $50 > 49$), the B term dominates. Precisely,

$$u_n = \left(\tfrac{5}{7}\right)^n \left[\, B + A\left(\tfrac{49}{50}\right)^n \right],$$

and $\left(\frac{49}{50}\right)^n \to 0$, so the bracket tends to B. If $0 < 6u_0 < v_0$ then $B = 3(6u_0 - v_0) < 0$, so the bracket is negative for all sufficiently large n. Since $\left(\frac{5}{7}\right)^n > 0$, u_n is negative for large n, as required. (Concretely, the bracket is negative once $|A|(\frac{49}{50})^n < |B|$.)

Physically, P is eventually knocked back away from the wall: Q has enough initial speed relative to P to keep driving it back, even though Q is much lighter.

Checks

- Take $e = 1$, $k = 1$ (equal masses, perfectly elastic): $(*)$ becomes $2u_n + 2u_{n-2} = 0$, so $u_n = -u_{n-2}$. Indeed equal masses colliding elastically swap velocities, so $u_1 = -v_0$ and $u_2 = -v_1$; then Q carries P's old velocity u_0 to the wall and back, giving $u_2 = -u_0$.
- At $n = 0$ the formula gives $A + B = u_0$, and at $n = 1$ it gives

  $$\tfrac{7}{10}(3v_0 - 17u_0) + \tfrac{5}{7}(18u_0 - 3v_0) = \tfrac{1}{70}(67u_0 - 3v_0),$$

  as it should.
- With $u_0 = 1$, $v_0 = 7$ we get $A = 4$, $B = -3$. Stepping through the collisions one at a time confirms the formula exactly, and u_n first becomes negative at $n = 15$, where $u_{15} \approx -0.00029$.

Common slips

- Forgetting that Q's velocity just before a collision is $-v_{n-1}$, not v_{n-1}: the wall reverses it.

- Getting the sign of the separation speed wrong in the restitution equation; after the collision Q is ahead of P in the positive direction, so it is $v_n - u_n$.
- Using $u_1 = \frac{7}{10}A + \frac{5}{7}B$ without having found u_1 from the collision equations; $(*)$ alone cannot give it.

Beyond STEP

The pair of equations (1) and (2) is a linear map taking the state (u_{n-1}, v_{n-1}) to (u_n, v_n), so the whole motion is repeated multiplication by a 2×2 matrix. The roots $\frac{7}{10}$ and $\frac{5}{7}$ are its eigenvalues, and long-term behaviour is governed by the largest one: this is the start of the theory of linear dynamical systems and difference equations. For more on the mechanics of collisions, see Kleppner and Kolenkow, *An Introduction to Mechanics*; for the matrix viewpoint, Strang, *Introduction to Linear Algebra*.

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